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Module Leader: Mr. Banuka Athuraliya

PhysioConnect

Group CS 113

Name	UOW & IIT No
Himaza Zahara Jayah	w20532374, 20222427
Lokukatagodage Sathnara Perera	w2052980, 20222129
Saad Mazhar	w20532233, 20221804
Sachira Delankawala	w20848480, 20230561
Thejan Paranagama	w2053789, 20222320
Azeezur Rahmaan Azwin	w2051763, 20230758

1. Abstract

This physiotherapy software offers athletes a comprehensive rehabilitation solution, including personalized exercise regimens, tutorials on critical first-aid methods, and the ability to assess injuries using a 2D body model. Through individualized injury assistance, it promotes resilience and motivation in athletes by providing progress tracking to help them follow healing milestones. The app helps athletes of all skill levels safely resume their sports and develop long-lasting health and strength by providing easily available advice on injury prevention and specialized care for various body parts. It is a special and vital resource for sports rehabilitation since it combines teaching, injury management, and recovery tracking.

2. Acknowledgment

I would like to express my heartfelt gratitude to Sir Banu, Sir Mithushan, and Sir Kushan for their invaluable support and guidance. Their constant feedback throughout the project has been instrumental in refining our ideas, overcoming challenges, and enhancing the quality of our work. Their dedication and insights have greatly contributed to our progress and motivated us to strive for excellence. Thank you for your mentorship and for always being there to help us achieve our goals.

3.Table of content

1. Abstract **2**
2. Acknowledgment **3**
3.Table of content 4
4. Problem **5**
4.1 Introduction to the problem 5
4.1.1 High costs 5
4.1.2 Time consumption 5
4.1.3 Efficiency Issues 6
4.1.4 Non-awareness of home remedies and real-time care 6
4.2 Problem background and stats 6
4.3 Examples in the problem 9
4.4 Attempted solutions of the Competitors 9
5.Solution **11**
5.1 Key Objective 11
5.2 Vision 11
5.3 Scope 11
6. Target Audience **12**
6.1 Primary Audience 12
6.2 Secondary Audience 12
7. Resource Requirements **13**
8. Features of the Solution **14**
8.1 AR implementation 14
8. 2 Customized Injury Remedies 14
8.3 Video Consultations 14
8.4 Progress Tracking 15
8.5 Educational Resources 15
8.6 Pain Monitoring 15
8.7 Customizable Exercise Plans 15
8.8 Gamification 16
8.9 Electronic Personal Health Record 16
8.10 In-app Messaging 16
8.11 AI Chatbot for Physiotherapy Advice 17
8.12 Nearby Facilities Locator 17
8.13 One-Tap Emergency Contacts 17
8.14 2D Model 17
8.15 Customized Injury Remedies 18
8.16 User profile and injury history 18
8.17 Notifications and Reminders (Tip of the day) 19
9. References **20**
10. Workload Matrix **21**

4. Problem

4.1 Introduction to the problem

Physiotherapy is critical for helping people recover from accidents, manage chronic diseases, and enhance mobility. However, various limitations prevent physiotherapy's accessibility and efficiency in Sri Lanka and across the world. These obstacles include expensive expenses, large time requirements, and a lack of understanding of home-based therapies. Furthermore, access to real-time, tailored physiotherapy services remains a challenge, sometimes impeding ideal healing results. The physiotherapeutic recovery or the process itself takes a considerable amount of time.

4.1.1 High costs

Physiotherapy treatments are expensive, especially in areas where health insurance coverage is restricted. A single physiotherapy session in Sri Lanka, for example, might cost between LKR 3,000 and LKR 8,000 per session, Sessions in the United States typically cost $75 to $120, with fees ranging from $100 to $150 for specialist treatments like in-home therapy. For people who need many weekly appointments, the costs build up, restricting continuous access to vital care.

4.1.2 Time consumption

The physiotherapeutic recovery or the process itself takes a considerable amount of time so in search for a physiotherapist or sessions would be a waste of time for a simple cure process of such kind. Physiotherapy frequently necessitates many trips to clinics or hospitals, with each session lasting 45 minutes to an hour or longer. For patients with hectic schedules or restricted mobility, this frequent time commitment becomes unrealistic, compromising adherence to treatment procedures and impeding recovery and inconsistency session plan which led to severe issues with the injury getting worse.

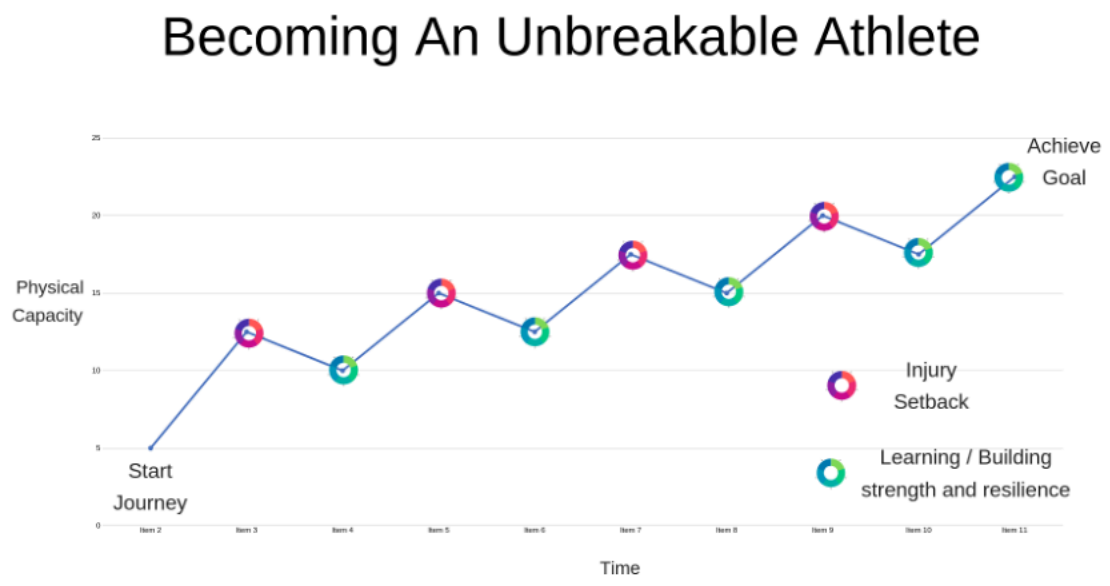
4.1.3 Efficiency Issues

Patients may not obtain the most effective treatment if they do not have constant access to high-quality care. Efficiency is especially impaired when patients do not complete specified exercises or do not have frequent access to a physiotherapist for continued coaching, which can cause delays in recovery.

4.1.4 Non-awareness of home remedies and real-time care

Many patients are unaware of excellent home-based therapies and activities that may be performed independently to aid in their recovery which will save them their money, time and effort for physical sessions and consultations. Furthermore, access to real-time feedback and instruction for physiotherapy care at home remains a barrier, leaving patients unclear of their progress or technique.

4.2 Problem background and stats



By assisting athletes in transforming injury setbacks into chances for growth and resilience, this app empowers them. It offers focused resources like individualized injury tracking, recovery activities, and useful tutorials that give athletes the skills and information they need to handle recovery with assurance. Athletes may readily access information specific to their own recovery path with tools like progress

monitoring, first-aid tutorials, and a 2D body model for pinpointing injury spots. This method helps athletes recover more quickly and develop a proactive mindset, which makes it simpler for them to avoid injuries in the future and develop long-lasting resilience.By assisting athletes in transforming injury setbacks into chances for growth and resilience, this app empowers them.

			Number of injuries by age			
Sport, activity or equipment	Injuries (1)	Younger than 5	5 to 14	15 to 24	25 to 64	65 and older
Exercise, exercise equipment	445,642	6,662	36,769	91,013	229,640	81,558
Bicycles and accessories	405,411	13,297	91,089	50,863	195,030	55,132
Basketball	313,924	1,216	109,696	143,773	57,413	1,825
Football	265,747	581	145,499	100,760	18,527	381
ATV's, mopeds, minibikes, etc.	242,347	3,688	42,069	61,065	122,941	12,584
Skateboards, scooters, hoverboards	230,506	8,513	64,439	61,546	88,328	7,680
Playground equipment	208,257	53,585	136,267	6,978	9,740	1,687
Swimming, pools, equipment	187,465	21,893	80,356	28,091	43,604	13,521
Soccer	179,284	1,313	80,540	68,602	27,997	832
Baseball, softball	136,874	2,560	61,336	39,981	29,232	3,765
Trampolines	123,014	32,186	70,400	10,805	9,366	257
Skating (excl. In-line)	76,505	356	31,816	14,117	28,278	1,938
Lacrosse, rugby, misc. ball games	66,882	455	20,252	20,142	15,334	10,699
Snowskiing, snowboarding	66,633	173	12,493	24,298	26,267	3,402
Golf	60,897	1,729	9,166	5,080	21,108	23,815
Fishing	57,257	1,204	10,711	8,169	27,733	9,439
Volleyball	50,643	82	16,923	25,461	7,151	1,027
Horseback	46,577	644	8,462	10,003	22,958	4,511
Hockey	37,893	186	10,967	16,469	9,494	778
Martial arts	30,113	22	9,288	8,336	11,969	496
Beach, picnic, camping equipment	25,935	2,144	3,900	2,624	11,868	5,399
Track and field activities, equipment	22,591	78	9,504	9,413	3,180	416
Racquet sports	21,730	196	2,901	3,627	7,384	7,621
Water skiing, tubing, surfing	21,180	81	2,792	6,207	10,969	1,131
Boxing	17,445	63	2,017	8,545	6,784	36
Toboggans, sleds, snow discs, etc.	14,040	1,436	6,136	2,635	3,261	572
Nonpowder guns, BB'S, pellets	11,603	519	3,286	3,443	3,869	487

(1) Treated in hospital emergency departments.

Source: National Safety Council analysis of U.S. Consumer Product Safety Commission NEISS data. National Safety Council. Injury Facts®.

Athletes of all ages suffer injuries in a variety of activities, ranging from injuries due to overuse in grownups to sprains in children from high-contact sports. The need for easily accessible injury support services across age groups and sports disciplines is highlighted by the necessity of customized rehabilitation and preventive care in managing age-specific recovery needs.

Many body parts are affected by sports injuries, with certain body parts being more vulnerable to particular injuries than others. The most often injured areas are the knees, shoulders, ankles, and wrists, each of which is linked to a certain sport, according to data from HHS. For example, shoulder injuries are common in overhead sports like tennis and swimming, while ankle sprains are common in sports like basketball that require rapid direction changes. The necessity for a platform that provides specialized treatment and prevention advice for various body parts is highlighted by the variety of injury sites.



With 50% of sports injuries considered preventable, high school athletes in particular could greatly benefit from targeted physiotherapy support after injuries occur. Core injuries alone account for 10% of these cases, impacting performance and increasing the need for rehabilitation tools. Given the athletic demands of popular sports like football, track, and volleyball, injuries from repetitive motions (throwing, kicking, running) are common. After these injuries, athletes often need structured recovery guidance to ensure safe healing and prevent future setbacks, underscoring the value of an accessible physiotherapy app to support their journey back to peak performance.

4.3 Examples in the problem

Imagine Alex, a high school cricket player, who frequently experiences minor injuries while playing.During a recent match, he injured his ankle, causing pain and discomfort. However, with an important game approaching, Alex needs a quick and effective solution to manage the injury so he can continue playing, rather than taking time off to visit a hospital.He requires a tool that provides immediate guidance on how to apply KT-tape correctly to the affected muscle, as applying the tape will help stop the spread of damage to the surrounding tissue. Additionally, he needs quick exercises and stretches to alleviate discomfort, enabling him to manage his injury effectively on the field.

4.4 Attempted solutions of the Competitors

Feature	Omada	Physitrack	SimpleTherapy	Our Product (Physio connect)
Video Consultations				
Progress Tracking				
2D Model Tutorials				
Pain Monitoring				
In-app Messaging				
Gamification				
EPHR (Electronic Personal Health Record)				

AI Chatbot for Physiotherapy Advice				
Daily Tip Generator				
Nearby Facilities Locator				
Customizable Exercise Plans				
One-Tap Emergency Contacts				
Educational Resources				
AR-based Physiotherapy Sessions				

YES

NO

What makes our product differ from the other competitors in the market

Omada: A software application designed specifically for managing patients with chronic diseases including but not limited to diabetes and hypertension. The app provides features that promote a healthy lifestyle among its users ranging from personal coaching through goal setting and advice on behavior change.

SimpleTherapy : A pain management focused, at-home physical therapy application. It provides pain specific self-directed, evolving exercise plans- for instance, back/knee pain where the plan changes depending on the user's response

Physitrack : A remote patient care monitoring platform for physiotherapists. The functional features include a vast content library of exercise demonstration videos, metrics of progress evaluation and means of communicating which satisfy HIPAA requirements for use in professional physiotherapy settings

Our product : Factors that distinguish Physio Connect(our product) from Omada, SimpleTherapy, and Physitrack include the advanced interactivity and personalization it offers, which includes such tools as the 2D muscle model for visualization of injuries and the geo-location tool for increased accessibility and patient participation. Also, it combines an effective EHR system and elaborate emergency instructions, making it one of the best resources for short and long-term physiotherapy care. This ensures health care delivery is extended and focused on the patient, unlike other competitors out there.

5.Solution

5.1 Key Objective

The major goal is to develop an accessible, cost-effective, and time-saving virtual platform for physiotherapy that overcomes present barriers to access, knowledge, and ongoing care. This program intends to democratize physiotherapy by making critical treatments available electronically to people all around Sri Lanka and beyond, while also offering real-time, individualized assistance to enhance recovery results.

5.2 Vision

The goal is to establish an easily accessible virtual physiotherapy platform that allows people to control their own rehabilitation. We hope to increase patient autonomy, treatment adherence, and proactive self-care by utilizing digital technologies, making physiotherapy accessible to everybody, regardless of geography or financial status.

5.3 Scope

This platform will provide a full digital solution for users at all phases of their physiotherapy experience. It will meet a wide range of demands, such as initial evaluations, frequent exercise instruction, progress tracking, and real-time assistance. The scope includes:

- Providing virtual consultations with qualified physiotherapists.
- Educating people about successful home-based treatments and activities.
- Tracking recovery progress to provide individualized advice.

- Reducing the time and financial obstacles to receiving physiotherapy care.
- Gamification, AI-based guidance, and ongoing assistance are all used to increase user engagement.

6. Target Audience

6.1 Primary Audience

Our primary audience comprises athletes and sports enthusiasts who are highly active and frequently face injuries due to the physical demands of their sports and fitness routines. Their injuries often stem from intense physical activity, repetitive strain, and high-impact sports movements. By focusing on this group, we aim to enhance their recovery process, help them perform exercises safely, and reduce the risk of re-injury, ultimately supporting them in achieving their peak performance.

6.2 Secondary Audience

Fitness fanatics, computer users, authors, accident victims, senior citizens, and office professionals are among our secondary audience members; each category faces difficulties and dangers of harm. For example, extended sitting and poor posture are common causes of neck, shoulder, and back discomfort among office workers, writers, and computer users. Meanwhile, accident victims and elderly adults may struggle with recovery and mobility issues. Regardless of age or physical restrictions, our goal is to provide solutions that meet the needs of these groups to help them recover more quickly, maintain an active, healthy lifestyle, and improve their posture.

7. Resource Requirements

Wearable Devices (Optional): Optional integration with wearable sensors for advanced features like biofeedback, motion tracking, and posture correction.

Software Requirements

- **Operating System Compatibility**: The platform will be compatible with iOS, Android, and web-based operating systems to ensure accessibility across various devices.
- **Database Management:** A robust and secure database system to store user data, exercise programs, and patient-therapist communication records.
- **Video Processing Software**: For remote sessions and real-time exercise monitoring; it should include compression and optimization capabilities to support smooth streaming.
- **User Interface and Experience (UI/UX) Tools**: Software to design an intuitive interface for easy navigation by patients, including accessibility features.
- **Security Tools**: End-to-end encryption, secure login, data encryption, and compliance with data protection standards (e.g., HIPAA) for safeguarding sensitive health data.

Technology Stack

Frontend Development
Languages: HTML, CSS, JavaScript
Frameworks/Libraries: React.js (for web), React Native or Flutter (for cross-platform mobile app)
Backend Development:

- Languages: Node.js
- Frameworks: Express.js (for Node.js) or Django (for Python)
- Database: MongoDB or PostgreSQL
- Algorithms: Computer vision for movement analysis and NLP for chatbot features

8. Features of the Solution

8.1 AR implementation

The Augmented Reality (AR) feature provides a virtual physiotherapy assistant that guides users through exercises with 3D visual aids. Using a mobile device's camera, patients can follow on-screen models to ensure correct posture and alignment. The AR system tracks movements in real time and offers immediate feedback, allowing patients to self-correct and prevent injury. This interactive approach increases engagement, enhances exercise accuracy, and makes remote physiotherapy accessible and effective.

8. 2 Customized Injury Remedies

These remedies consider factors such as the user's type of injury, acuteness, wounded area, and the progress in healing to provide personalized treatment plans or exercises tailored to an individual's recovery needs. Through customization, users gain access to K-taping tutorials, tailored exercises, targeted stretches, and comprehensive instructions on oil treatments. It enables users to perform recovery routines at home using common household items, facilitating them to conduct therapeutic exercises without needing direct professional guidance. This approach is more likely to support recovery by utilizing personal health data rather than providing generic recommendations, enabling users to progress more effectively and safely.

8.3 Video Consultations

Video Consultations allow patients to interact with physiotherapists remotely in order to assess, advise or modify treatment plans. This is a convenient and flexible feature, especially for patients who cannot move or have to travel a long distance, thus improving accessibility and engagement with patients.

8.4 Progress Tracking

Athletes can track changes in strength, flexibility, and pain levels over time by using the Progress Tracking function to keep tabs on their rehabilitation process. The users can celebrate successes, set recovery objectives, and modify routines as necessary with the help of visible indications and frequent updates. By offering a clear route back to optimal performance and guaranteeing that users stick to their recovery plan, this element increases incentive. Athletes and their therapists might learn useful tactics to assist them avoid re-injury and strengthen resilience-building techniques by gathering this data.

8.5 Educational Resources

To help users comprehend their injuries, the healing process, and preventative measures, the app offers extensive educational resources. This includes detailed information on various injuries, their causes, symptoms, and treatment options. Users have access to educational videos, articles, and interactive resources that guide them through stretches, exercises, and recovery methods, enabling people to make informative decisions regarding their recovery journey.

8.6 Pain Monitoring

The Pain Monitoring function enables users to measure their pain levels over time, assisting them in identifying trends and triggers associated with their injury. Users may contribute crucial information to their healthcare professionals by documenting pain intensity, location, and any accompanying symptoms. This allows for more individualized treatment strategies. This feature not only increases self-awareness, but also enables people to better control their discomfort, encouraging proactive participation in their recovery path.

8.7 Customizable Exercise Plans

Using information supplied by users, the app generates highly customized workout plans based on each user's requirements and injury history. These plans are designed to help users recover effectively, improve strength and flexibility, and prevent future injuries. The software creates

personalized exercise regimens that are safe, efficient, and inspiring by considering factors like the degree of injury, fitness level, and particular objectives. To maximize their recuperation process, users can effortlessly monitor their progress and modify their plans as necessary.

8.8 Gamification

The Gamification feature introduces interactive elements, such as rewards, challenges, and progress milestones, to enhance user engagement and motivation. By turning routine exercises into enjoyable activities, users are encouraged to stay consistent in their recovery journey. Points, badges, and levels can be earned for completing daily exercises and meeting goals, creating a sense of accomplishment and progress. This approach not only makes physiotherapy sessions more engaging but also reinforces adherence, promoting better outcomes.

8.9 Electronic Personal Health Record

The Electronic Personal Health Record (EPHR) system allows users to securely store, view, and manage their health data in one convenient location. It provides a comprehensive record of past injuries, treatments, exercise plans, and progress reports, all accessible to both patients and therapists. By keeping all health-related information centralized and organized, the EPHR system enables better decision-making and continuity of care, enhancing the overall therapeutic experience.

8.10 In-app Messaging

In-App Messaging function helps the various parties such as the patients and physiotherapists to communicate directly through the app ensuring that there are quick and safe exchanges regarding the progress of the treatment, queries about the treatment, and the changes in the treatment. This feature ensures that there is always support given to the patient even in between the appointments encouraging compliance to the therapy and providing a hassle-free way of addressing the patient’s concerns instantly.

8.11 AI Chatbot for Physiotherapy Advice

The software incorporates a chatbot driven by AI to offer real-time physiotherapy guidance and assistance. Users can ask questions about their injuries, recovery process, or exercise routines, and receive immediate, evidence-based responses from the chatbot. Throughout the healing process, this assistant can offer personalized advice, fix typical problems, and reassure. Through the utilization of AI, the application enables users to actively participate in their recovery and attain the best possible outcomes.

8.12 Nearby Facilities Locator

To assist users in locating medical professionals, physical therapists, and other pertinent services in their area, the app includes a nearby facility locator. Users may easily find and get in touch with professionals who can assist with their rehabilitation and recovery thanks to this feature. The software makes it convenient to discover certified healthcare providers and guarantees that users receive timely and appropriate care by making vital services easily accessible.

8.13 One-Tap Emergency Contacts

The One-Tap Emergency Contacts functionality allows customers to swiftly receive vital assistance during emergency scenarios. Individuals may connect to pre-selected emergency contacts, such as family members, friends, or medical professionals, with a single touch, offering peace of mind and allowing for immediate aid when needed. This function improves user safety by reducing communication barriers during crises, allowing for fast assistance in the event of serious injuries or health issues.

8.14 2D Model

A key component of the program is the 2D Model tool, which enables athletes to identify the location of their ailment by only picking a region on an intricate body diagram. Even those without medical training may more properly convey pain spots and damage details with this graphic technique. The software offers customized recommendations, including first-treatment

protocols, area-specific exercises, and first-aid lectures, based on the region that has been selected. By providing users with an easy-to-use, interactive method to recognize, comprehend, and treat their ailments, this feature expedites the road to a successful recovery.

8.15 Customized Injury Remedies

Safe and efficient treatment alternatives are provided by the Customized condition Remedies feature, which creates recovery regimens specifically suited to each athlete's condition and body part. This feature recommends easy fixes with easily accessible materials, like cold compression instruments for the home, guided stretching exercises, and strength-training activities targeted at injured joints or muscles. Furthermore, it offers instructions on how to use first-aid supplies like Kinesiology (K) tape, assisting athletes in learning self-care methods for both short-term assistance and long-term recuperation. This tool offers a useful, hands-on approach to injury management by customizing guidance to each user's circumstances, allowing athletes to make consistent, knowledgeable progress toward recovery.

8.16 User profile and injury history

The User Profile and Injury History function of the Physio Connect provides users with the ability to create detailed profiles that encompass information about themselves, their injuries as well as their past medical histories, treatment goals, and monitors the improvement within a certain time frame making for a personalized and effective recovery process.

Through the User Profile and Injury History feature, patients are able to track their own recovery progression, thus improving their compliance to treatment and allowing physiotherapists to make informed choices and adjust treatment plans, ultimately providing the same care even when the patient changes their therapist

8.17 Notifications and Reminders (Tip of the day)

This feature offers notification and reminders to keep users engaged and motivated and to remind the time to get medicines and treatments for injuries. This app sends a notification named "Tip of the Day" to provide useful advice daily such as injury prevention, wellness habits, reminders about exercises, motivation quotes.

8.18 First Aid tutorials

The First Aid Tutorials feature functions as a full virtual toolkit, providing users with DIY learning materials for proper injury care. It offers thorough lessons on how to use kt tape and other therapeutic tapes, as well as advice on massage and pain relief techniques and exercises for sprains. These kinesiology lessons allow users to execute key physiotherapy activities from the comfort of their own homes or on the road. By providing individuals with practical skills, this feature improves their capacity to treat minor injuries on their own and promotes a proactive attitude to their health and rehabilitation.

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10. Workload Matrix

TASK	Himaza	Sachira	Sathnara	Azeez	Thejan	Saad
1.Problem						
1.1 Introduction to problem (Detailed explanation about the problem identified including stats to justify why it is a problem)						
1.1.1 Problem background and stats (Explain problem background using stats)						
1.1.2 Examples in the problem (Describe your problem using an example scenario)						
1.1.3 Attempted solutions of the Competitors (with a Feature comparison chart - Example)				chart - All*		
2.Proposed Solution						
Key objective and vision						
Scope						
3.Target Audience						
Primary and secondary audiences						
4. Resource Requirements						
Hardware, Software, Technology Stack						
5. Features of the solution						
- Interactive 2D Model						
- Customized Injury Remedies						
- First Aid Tutorials						
- User Profile and Injury History						
- Notifications and Reminders (Tip of the day)						
- Ar iplementation						
Video Consultations - A						
Progress Tracking - H						
Educational Resources - SATH						
Pain Monitoring - SACHI						
Customizable Exercise Plans - T						
Gamification - SAAD						
Electronic Personal Health Record - SAAD						
In-app Messaging - A						
AI Chatbot for Physiotherapy Advice - SATH						
Nearby Facilities Locator - T						
One-Tap Emergency Contacts - SACHI						